FLOODGUARD TUMANA BRIDGE

Complete Defense Record and Evidence Guide

R3.1 documentation repair | one-day-ahead OLS Multiple Linear Regression time-series model

Item	Verified position
Target	PAGASA-reported daily Tumana water-level observation
Deployed model	Candidate 8: Tumana_WL(t-1) + PAGASA Science Garden Rain(t-1)
Method	Transparent OLS Multiple Linear Regression; K = 2; exact calendar lags
Model purpose	Next-day daily monitoring decision support only
Final status	TUMANA_MLR_TIME_SERIES_MODEL_PASS

Plain-language answer: FloodGuard forecasts the next PAGASA-reported daily Tumana water-level observation. It does not claim to forecast an hourly peak, a daily maximum, flood depth, flood probability, or an automatic evacuation trigger.

How to use this record

• Read Sections 1-4 to understand the model and why it was chosen.

• Use Sections 5-8 for the evidence: lags, screening, validation, persistence, and availability.

• Use Sections 9-12 to prepare concise answers for the panel and to explain the system safely.

1. What Tumana Forecasts

Tumana is handled differently from Sto. Nino and Nangka because its historical PAGASA source provides one water-level value per calendar day. The file does not confirm whether that value is a daily maximum, daily mean, fixed-time reading, or another within-day observation. We preserved each value exactly as supplied.

Question	Answer
Exact target name	PAGASA-reported daily Tumana water-level observation
Unit	meters (m)
Forecast horizon	next calendar day
What we did not do	No synthetic hourly records, no 24/24 rule, no daily max/mean transformation, no imputation
Safety boundary	Daily monitoring decision support; not an intraday peak forecast or sole emergency trigger

Why this is still valid

A daily source can support a daily time-series model when the target is named honestly. The model forecasts the source-reported value for the next day. It must not be relabeled as a peak-stage or threshold-alert model.

> **Defense line: “We accepted the daily source cadence as a documented constraint. We did not invent hourly data or assign unverified measurement semantics.”**

2. The Required OLS MLR Time-Series Method

Ordinary Least Squares (OLS) estimates a visible equation by choosing coefficients that minimize squared prediction errors. Multiple Linear Regression (MLR) requires at least two predictors. Time-series forecasting requires past information only.

Final production equation

Tumana_WL(t) = 1.514724 + 0.873544 x Tumana_WL(t-1) + 0.008482 x PAGASA_SG_Rain(t-1)

Term	Meaning
Tumana_WL(t-1)	The prior day PAGASA-reported Tumana observation. This is the time-series memory term.
PAGASA_SG_Rain(t-1)	Prior-day daily rainfall at PAGASA Science Garden, in millimeters.
K = 2	Two predictors: this is MLR, not simple linear regression.

Plain-language interpretation

• Yesterday's river level is the strongest memory signal for tomorrow's reported level.

• The rainfall term provides additional information about rain-driven movement.

• A 10 mm increase in prior-day Science Garden rain corresponds to an estimated 0.0848 m increase in the next reported level, holding the prior Tumana value constant.

• This is an observed statistical association in this dataset, not a claim of universal physical causation.

3. Data, Exact Calendar Lags, and Splits

All lags were constructed on a continuous 2016-01-01 to 2025-12-31 calendar grid. A value for date t uses the record exactly one calendar day earlier, not the previous nonblank spreadsheet row. Missing values remain missing.

Period	Dates	Purpose
Training	2016-01-01 to 2022-05-26	Fit OLS coefficients and conduct HC3/VIF screening
Validation	2022-05-27 to 2023-12-31	Compare frozen candidates chronologically
Final refit	2016-01-01 to 2023-12-31	Create the production equation after selection
Retrospective	2024-01-01 to 2025-12-31	Availability and historical operational review; not an untouched holdout

Missing-data rules

• No zero-filling, forward-fill, backward-fill, interpolation, or made-up rainfall.

• Each candidate uses its own complete cases. Candidates are compared only on their shared validation dates.

• The Tumana series has a historical 2019-2021 outage period; calendar lags do not jump over those gaps.

> **Defense line: “Using the prior spreadsheet row would create false lags across missing periods. Our calendar grid prevents that leakage.”**

4. Why Candidate 8 Was Selected

All 15 candidates were frozen before validation. The project requires the deployed model to be an OLS MLR with at least two predictors and at least one water-level lag. Candidate 1 remains a useful AR1 benchmark but cannot be deployed because K = 1.

Candidate	Predictors	Governance result
C01	Tumana(t-1)	AR1 benchmark only: K = 1, not MLR
C03	Tumana(t-1) + Boso-Boso rain(t-1)	Ineligible: training HC3 rain p = 0.2183
C04	Tumana(t-1) + Campana rain(t-1)	Ineligible: training HC3 rain p = 0.1884
C08	Tumana(t-1) + SG rain(t-1)	Eligible and selected
C14	Tumana(t-1) + Tumana(t-3) + SG rain(t-1)	Eligible alternative

Frozen selection hierarchy

• MLR requirement: K >= 2 and at least one water-level lag.

• VIF must be <= 5.0.

• Every slope must have training-sample HC3 p < 0.05.

• Among eligible MLR candidates, choose by lower validation MAE and exact-common-date pairwise MAE comparison.

• Compare with true persistence separately; do not hide when persistence wins a metric.

5. HC3 and VIF: The Reliability Screens

OLS supplies the equation. HC3 robust standard errors test whether each added predictor remains reliably associated with the outcome even when ordinary and storm-day errors have different spread. VIF checks whether predictors overlap so strongly that coefficients become unstable.

Candidate	Training HC3 result	Max VIF	Decision
C08	WL $p = 7.38e-34$; rain $p = 4.88e-6$	1.12	Pass
C14	All slopes $p < 0.05$; maximum $p = 0.0460$	3.01	Pass
C04	Campana rain $p = 0.1884$	1.19	Fail HC3
C03	Boso-Boso rain $p = 0.2183$	1.19	Fail HC3

Simple explanation

• HC3 asks: “Does this predictor still look dependable after using a more cautious error calculation?”

• VIF asks: “Are the predictors too redundant with each other?”

• C08 passes both screens. Campana and Boso-Boso had good-looking scores in some comparisons, but their added rain effects did not pass the robust training test.

HC3 does not alter the OLS coefficients. It is a stricter check on the uncertainty around them.

6. Validation: Candidate 8 Versus Candidate 14

The two eligible MLR candidates make a real trade-off. The project froze MAE and pairwise MAE as the selection rule, so Candidate 8 was selected. Candidate 14's lower RMSE is still reported openly.

Metric	C08 selected	C14 eligible alternative	What it means
Validation MAE	0.1481 m	0.1514 m	C08 is lower on average absolute error
Validation RMSE	0.3244 m	0.3101 m	C14 is lower when large errors receive more weight
Eligible-only pairwise MAE	1 / 1 win	0 / 1 wins	C08 wins the frozen head-to-head rule
VIF	1.12	3.01	Both pass <= 5.0

Why both metrics matter

MAE treats a 0.10 m miss and a 1.00 m miss proportionally. RMSE squares errors first, so unusually large misses have greater influence. It is correct to say C08 was selected by the frozen MAE governance rule; it is not correct to say C08 is better than C14 on every metric.

> **Defense line: “We did not hide C14's lower RMSE. Our candidate rule was frozen before selection and prioritizes MAE plus exact-common-date pairwise MAE among eligible MLR models.”**

7. Candidate 8 Versus True Persistence

True persistence is not Candidate 1. Persistence simply repeats the prior observed level: prediction(t) = Tumana_WL(t-1). Candidate 1 is a fitted AR1 OLS equation with an intercept and fitted slope.

Model	Equation type	Validation MAE	Validation RMSE
True persistence	Tumana(t-1)	0.1416 m	0.3604 m
Candidate 1 AR1 OLS	0.487709 + 0.961230 x Tumana(t-1)	0.1442 m	0.3543 m
Candidate 8 OLS MLR	Tumana(t-1) + SG rain(t-1)	0.1481 m	0.3244 m

Honest conclusion

• Persistence has lower MAE on C08's exact validation dates, especially on quiet/baseflow days.

• C08 has lower RMSE: 0.3244 m versus 0.3604 m, a 9.99% reduction in squared-error-sensitive performance.

• The project retains persistence as both a benchmark and a runtime fallback. It is not hidden or treated as identical to AR1 OLS.

8. Final Refit Evidence

After Candidate 8 was selected using the chronological validation period, the model was refit using all available data from 2016-01-01 through 2023-12-31. Refit N = 1,648 and refit R-squared = 0.842641.

Parameter	Coefficient	HC3 SE	HC3 t	HC3 p	95% CI
Intercept	1.514724	0.768255	1.9716	0.0488	[0.0079, 3.0216]
Tumana(t-1)	0.873544	0.062491	13.9787	4.8294e-42	[0.7510, 0.9961]
SG rain(t-1)	0.008482	0.001481	5.7254	1.2242e-08	[0.0056, 0.0114]

Formula proof

The companion Excel workbook contains 7,167 live formulas and zero formula errors. The package also includes candidate tables, exact-calendar lag audit, persistence comparisons, HC3/VIF tables, prediction files, checksums, and a reproducibility script.

Important: the intercept is not a standalone water level or a physical river-bed elevation. It is the fitted constant term needed by the linear equation.

9. Deployment Logic and Availability

The production calculation uses C08 only when both required prior-day values are present. The Science Garden rainfall history ends after 2024, so the primary C08 model is not available for most 2025 retrospective dates.

Runtime condition	System output
Tumana(t-1) and SG rain(t-1) are valid	Use Candidate 8 OLS MLR equation
Tumana(t-1) valid; SG rain(t-1) missing	Use true persistence: prediction = Tumana(t-1)
Tumana(t-1) missing	Return forecast unavailable; do not impute

Availability truth

Measure, 2024-2025 retrospective	Evidence
Observed Tumana target days	566
C08 primary-model days	311 / 566 = 54.95%
Days with a usable prior Tumana level (MLR or persistence)	547 / 566 = 96.64%
Forecast unavailable because Tumana(t-1) missing	19 days

Correct claim: fallback improves coverage when rain is missing, but it does not create 100% coverage. The system has a forecast only when the prior-day Tumana observation exists.

10. Limitations and Safe Claims

Limitation	How FloodGuard handles it
Daily source semantics are unconfirmed	Uses the exact target name; does not call it max, mean, or fixed-time stage
No hourly Tumana source	Does not create hourly observations or an intraday peak forecast
Science Garden rain ends after 2024	Uses true persistence when rain is unavailable and reports primary availability
Prior Tumana value can be missing	Returns forecast unavailable; no filling or interpolation
Daily source may not match official warning thresholds	Does not map Tumana forecast to automatic alerts, sirens, or evacuation triggers

What we can say

• “Tumana is a station-specific daily OLS MLR time-series component.”

• “It forecasts the next PAGASA-reported daily Tumana observation.”

• “It supplements monitoring; live observed values and official warning protocols remain the safety authority.”

What we must not say

• “It predicts hourly flood peaks.”

• “It predicts flood depth or probability.”

• “It automatically determines evacuation status.”

11. Panel Questions and Short Answers

Likely question	Concise answer
Why use MLR?	It is a stated project requirement and it lets us quantify the added value of rainfall while keeping the equation transparent.
Why not use Campana or Boso-Boso?	They were tested, but their added rainfall slopes failed the training HC3 robust-significance rule. We did not select a predictor based only on a favorable score.
Why Science Garden rain?	It passed HC3 and VIF and, among eligible MLR candidates, C08 won the frozen MAE and pairwise-MAE selection rule.
Is C08 best in every metric?	No. C14 has lower RMSE, while C08 has lower MAE and wins the frozen eligible-only MAE comparison. We disclose the trade-off.
Is C01 persistence?	No. C01 is a fitted AR1 OLS regression. True persistence repeats the prior observation exactly.
Why is 2025 primary availability low?	Science Garden rain ends after 2024. The system uses true persistence when rain is missing and returns unavailable only when the prior Tumana value is missing.

Closing defense position: “We chose the simplest eligible OLS MLR time-series model under a frozen, auditable rule; preserved the source semantics honestly; and made fallback and availability limitations visible rather than hiding them.”

12. Team Defense Checklist

Before presenting, every member should be able to explain the following without reading numbers word-for-word.

Check	Team-ready explanation
Target	PAGASA-reported daily Tumana water-level observation, not a proven daily maximum.
Method	OLS MLR time-series: prior Tumana level plus prior-day Science Garden rainfall.
Eligibility	MLR K >= 2, training HC3 $p < 0.05$ for slopes, and VIF <= 5.
Selection	C08 wins the frozen eligible-MAE/pairwise rule; C14's lower RMSE is disclosed.
Benchmark	Persistence is separate and has lower MAE, while C08 has lower RMSE.
Deployment	C08 when rain exists; persistence if rain is missing; unavailable if the prior Tumana level is missing.
Safety	Daily decision support only; no automated threshold or evacuation claim.

Package evidence: formula-backed workbook, frozen candidate definitions, full OLS and HC3 tables, VIF audit, validation predictions, pairwise comparisons, persistence comparisons, runtime parity tests, raw-source inventory, manifest, checksum, and reproducibility script.

End of Tumana R3.1 Complete Defense Record